Superman Suit: A Photovoltaic Neuromuscular Safety Net for Off-World Habitats

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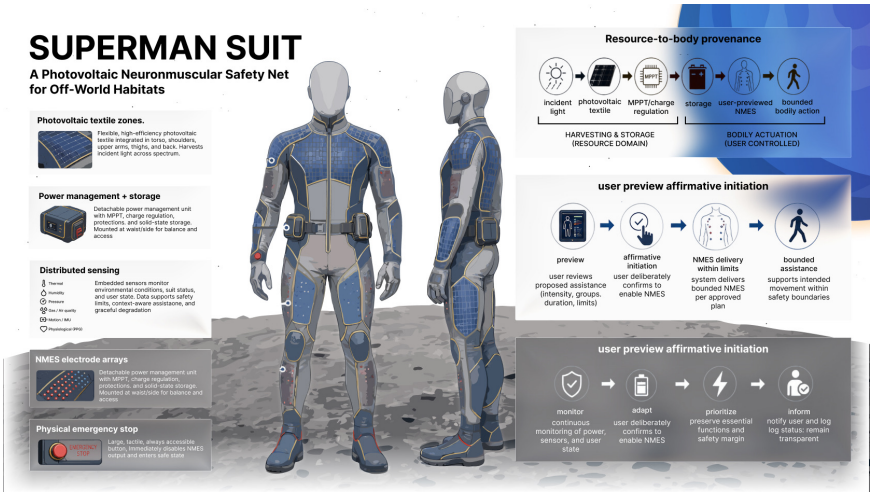


Fig. 1. Illustrative overview of the proposed Superman Suit as a resource-to-body system. The guide identifies the garment, photovoltaic zones, storage and control module, electrode locations, sensing, data/energy provenance, and wearer-facing safety controls.

Abstract. Future off-world habitats may treat the human body as another load on centralized systems for power, exercise, and health maintenance. This position paper argues that Space HCI should also investigate garments as distributed physiological infrastructure: body-proximate systems that harvest local resources, make scarcity legible, and preserve bounded care when centralized infrastructure is interrupted. We develop this position through the *Superman Suit*, a conceptual photovoltaic garment that stores harvested light energy and allocates it to user-controlled neuromuscular electrical stimulation (NMES) when ordinary exercise is temporarily unavailable. The suit is not proposed as an exercise replacement, clinical device, or flight-qualified system. Instead, it is a theory-bearing artifact for examining environmental intermittency, energy provenance, bodily automation, and crew agency. Following design science and Research through Design, we combine literature grounding, an annotated system model, failure-first interaction scenarios, and a bounded computational

energy probe. A 750,000-draw Monte Carlo probe under explicitly illustrative assumptions found robust energy support in deployed or strongly illuminated scenarios, conditional or assisted operation indoors, and predominantly assisted or unsupported operation during light interruption. We contribute the strong concept of a *metabolic safety net*, seven design propositions, and a staged research agenda that can challenge or falsify the proposal.

Keywords: Space HCI · wearable technology · neuromuscular electrical stimulation · photovoltaic textiles · design science · Research through Design · human autonomy

1 Introduction

Long-duration spaceflight makes bodily maintenance an infrastructure problem. Exercise countermeasures, clinical oversight, power, data, and maintenance are coupled to a habitat that is itself exposed to interruptions, constrained mass, and operational uncertainty. NASA's evidence review identifies reduced muscle mass, strength, endurance, and aerobic capacity as performance risks and treats exercise as a central countermeasure [1]. The practical burden is not only the exercise dose: setup, stowage, hygiene, scheduling, and equipment maintenance consume scarce crew time and habitat resources. Reports of cable wear in the International Space Station countermeasures system illustrate how a mundane component can interrupt an otherwise established capability [14,15].

This paper asks what Space HCI can contribute when a health garment is designed as infrastructure rather than as personal electronics. We propose the *Superman Suit*: a conceptual garment that harvests light through photovoltaic textile surfaces, stores a bounded energy reserve, and uses that reserve to support user-controlled NMES when a primary exercise capability is temporarily unavailable. The name is a memorable metaphor for a protective safety net, not a claim of superhuman enhancement.

Our position is deliberately narrower than a product claim. The suit is not an exercise replacement, a prescription for stimulation parameters, a clinical intervention, or a flight-ready system. It is a theory-bearing artifact through which to examine how environmental resources become bodily action. The central claim is:

> Space HCI should treat off-world health garments as distributed, resource-aware physiological infrastructure: systems that sense local resource availability, preserve a bounded reserve, explain how that reserve is allocated to the body, and maintain limited care during infrastructural interruption.

The paper makes three contributions. First, it introduces the *metabolic safety net*, a strong concept for limited physiological continuity during failure. Second, it instantiates that concept in an annotated suit architecture and interaction

model. Third, it contributes a design-science research agenda, including design propositions, failure conditions, and a computational probe that disciplines material plausibility without being mistaken for human validation.

2 Why This Is a Space HCI Problem

2.1 From exercise equipment to distributed infrastructure

Human spaceflight countermeasures are usually framed as dedicated equipment, scheduled procedures, or clinical protocols. This framing is useful for accountability, but it can obscure the relationship between environmental resources, maintenance work, bodily agency, and system failure. A garment makes these relationships intimate: energy is gathered on the body, stored near the body, and ultimately converted into a bodily stimulus. The design therefore has to communicate not just status, but provenance: where the energy came from, how much is available, how uncertain that estimate is, and what bodily action it can support.

The proposal builds on three established observations. First, exercise remains the benchmark countermeasure for deconditioning in spaceflight [1]. Second, NMES can evoke muscle contraction and has been investigated in spaceflight and bed-rest contexts, but its delivered current, monitoring, and resultant force require explicit measurement [3,4]. Third, photovoltaic textiles and flexible solar materials demonstrate a direction for body-conformal harvesting, while their practical performance depends strongly on geometry, illumination, packaging, and durability [6,7]. These observations support a research opportunity, not a deployment conclusion.

2.2 Why intermittency matters

The lunar south pole combines long-duration illumination geometry, shadow, thermal extremes, and dust concerns [8]. Mars adds atmospheric dust and regional or global storms that can attenuate sunlight [9]. Habitats also create their own interruptions: a solar-facing window may be unavailable, an external array may be isolated, or an exercise device may be reserved, degraded, or under maintenance. A resource-aware garment must therefore have explicit states for autonomous support, hybrid support, and unavailable support. A design that silently assumes continuous light converts a contingency system into a hidden single point of failure.

2.3 HCI stakes: agency, trust, and bodily governance

The suit would not merely display a battery percentage. It would negotiate a boundary between crew autonomy and automated resource allocation. A system that actuates the body must be previewable, interruptible, and intelligible. NASA's human-system standard provides a relevant human-factors baseline for

crew interfaces [10], while the safety and essential-performance requirements for nerve and muscle stimulators provide a reminder that a wearable concept cannot bypass medical-device constraints [13]. These requirements motivate the design concepts of resource-to-body legibility, bounded bodily automation, and safe scarcity developed below.

3 Design-Science Position

3.1 Design science and Research through Design

Design science develops knowledge by constructing and evaluating artifacts that extend human or organizational capabilities [17]. Research through Design (RtD) uses the act of making a designed artifact to produce knowledge that is relevant, inventive, and extensible beyond the artifact itself [18]. RtD knowledge may be provisional and contingent, but it remains rigorous when the artifact, design reasoning, annotations, and limits are made visible [19]. Strong concepts provide an intermediate level of abstraction: more transferable than a single product, but close enough to interaction to guide design decisions [20].

The Superman Suit is therefore treated as a *theory-bearing artifact*. It is not evidence that a photovoltaic NMES suit preserves muscle in space. Its role is to make a set of relationships discussable and falsifiable:

- environmental energy becomes a bodily reserve;
- the reserve is intermittent, uncertain, and physically situated;
- the user must understand and authorize bodily actuation; and
- a failure mode should degrade capability visibly rather than increase risk.

The design-science cycle has three linked parts. The relevance cycle frames exercise interruption, maintenance burden, resource scarcity, and crew autonomy. The rigor cycle draws on space-health evidence, NMES limitations, photovoltaic textiles, human factors, and medical-device safety. The design cycle constructs, annotates, stress-tests, and revises the suit concept. The output is intermediate-level design knowledge rather than a validated biomedical result.

3.2 The metabolic safety net

The central concept is a *metabolic safety net*: a secondary, distributed system that maintains a bounded physiological-care function when a primary countermeasure is interrupted. The word *bounded* is essential. The safety net does not need to reproduce every benefit of a treadmill, cycle ergometer, or resistive exercise device. It must instead preserve a small, well-defined capability while the habitat restores or replaces the primary system.

This concept produces four design commitments. First, contingency takes priority over substitution: the system is judged by continuity during interruption, not equivalence during nominal operation. Second, resource provenance precedes actuation: the wearer sees how an environmental resource is becoming bodily

action. Third, scarcity is safe: declining energy reduces or postpones capability and never justifies exceeding an approved stimulation envelope. Fourth, bodily sovereignty is explicit: sessions require preview, affirmative initiation, and an immediate local stop.

3.3 Design propositions

The following propositions are intentionally contestable. They provide intermediate-level claims that can organize future design and empirical work.

DP1 – Distributed continuity. Off-world health systems should preserve a minimum distributed care capability so bodily maintenance does not disappear completely when centralized equipment fails.

DP2 – Provenance before actuation. A resource-aware wearable should disclose energy origin, available quantity, uncertainty, and the bodily action that the reserve can support before actuation.

DP3 – Bodily sovereignty. Any system that actuates the body should require preview, affirmative initiation, immediate local stop, and intelligible post-action accounting.

DP4 – Safe scarcity. Energy scarcity should reduce, defer, or cancel functionality; it must never justify exceeding an approved stimulation or discomfort envelope.

DP5 – Countermeasure plurality. A physiological wearable should be one layer in a portfolio of countermeasures, not a totalizing replacement for exercise, clinical care, or habitat infrastructure.

DP6 – Failure-first evaluation. Resource-aware physiological systems should be evaluated through outages, uncertainty, sensor disagreement, and user refusal, not only through nominal operation.

DP7 – Evidence-bound speculation. Speculative artifacts become scientifically useful when each claim is labeled as established evidence, design inference, computational result, or future empirical question.

4 The Superman Suit as a Design Instantiation

4.1 System architecture

The proposed garment is a layered system rather than a single fabric. A photovoltaic textile layer collects incident light over torso and limb surfaces. A power-management layer performs maximum-power-point tracking, charge regulation, and electrical isolation. A storage layer holds a reserve sized for an explicitly bounded session. A sensing layer estimates illumination, garment contact, skin or electrode impedance, motion, temperature, and electrical output. A stimulation layer delivers a charge-balanced, clinician-defined NMES waveform through localized electrodes. A wearer-facing interface communicates provenance, reserve, uncertainty, session preview, and stop state. These layers are conceptual requirements, not a completed bill of materials.

The energy path is intentionally visible:

$$E_{\text{light}} \rightarrow E_{\text{PV}} \rightarrow E_{\text{stored}} \rightarrow E_{\text{NMES}} \rightarrow \text{bounded bodily action.} \quad (1)$$

At every transition, the system should expose losses and uncertainty. A high battery state cannot be presented as available stimulation energy if electrode impedance, converter overhead, thermal constraints, or session policy prevent delivery.

4.2 Interaction states

The garment has five user-visible states:

1. **Harvesting:** light and charge rate are displayed, with the active harvesting surface and environmental uncertainty.
2. **Reserve:** stored energy is reported as usable session time under the selected protocol, not only as a percentage.
3. **Preview:** the wearer or an authorized clinician sees the intended muscles, duration, waveform envelope, expected discomfort, and energy cost.
4. **Active:** the session shows delivered output, contact quality, stop affordance, and remaining reserve.
5. **Degraded or unavailable:** the system explains why a session is shortened, deferred, or blocked and offers non-actuating alternatives.

The design forbids autonomous background stimulation. A scheduler may recommend a session when resource and mission conditions are favorable, but the session must still require affirmative initiation. The emergency stop is physical, locally reachable, and independent of the display or network. If sensing disagrees or the wearer refuses, the safe outcome is to stop or defer.

4.3 Illustrative suit design

This section provides an annotated design guide for the author's later refinement. Figure 2 separates the photovoltaic and power layers from the bodily sensing and actuation layers. It annotates the photovoltaic zones, removable electronics, electrode and sensor locations, charging and data paths, and the physical emergency stop. It also shows what the wearer can see before and during a session. The most important visual distinction is between the *resource layer* (harvesting, storage, and uncertainty) and the *bodily layer* (sensing, stimulation, and user control).

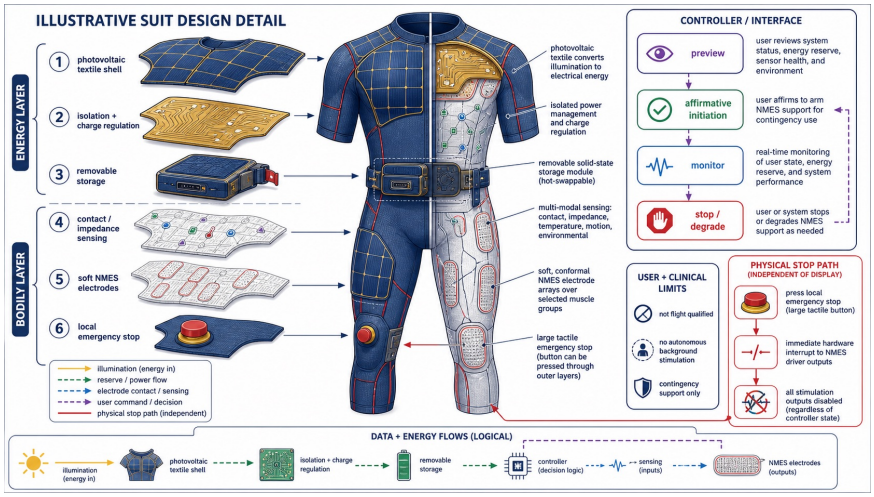


Fig. 2. Illustrative suit-design detail guide. The resource-to-body loop, safety boundary, and independent physical stop path are made legible at a glance.

5 Failure-First Scenarios and Safety Boundaries

The concept is evaluated by the situations in which its assumptions fail. Table 1 summarizes five scenarios used to structure the design probe and the interaction requirements.

Table 1. Failure-first scenarios for the Superman Suit.

Scenario	Resource condition	Required system response
Indoor habitat	Low-to-moderate, variable illumination	Show conditional support; recommend external charging or primary exercise.
Window-filtered habitat	Stronger but filtered illumination	Report conversion loss and usable charge duration.
Lunar deployment	High incident light but thermal, dust, and geometry constraints	Separate local harvest from mission-wide power and maintenance assumptions.
Mars clear deployment	Strong outdoor light with dust uncertainty	Make environmental confidence and predicted reserve visible.
Light interruption	Zero or near-zero harvest during a session	Stop, shorten, or switch to stored-reserve mode without increasing stimulation risk.

The safety boundary is not an optimization target. The control policy is lexicographic: (1) stop on unsafe contact, impedance, temperature, or command disagreement; (2) enforce the clinician-defined stimulation envelope; (3) preserve a minimum reserve for emergency and diagnostic operation; (4) optimize session timing and harvested-energy use. An energy-aware controller may choose a smaller session, but it may never choose a higher current merely because more physiological work is desired.

6 A Bounded Computational Design Probe

6.1 Purpose and scope

The computational study is an evaluative design probe. It asks whether the proposed resource-to-body loop is materially plausible under declared assumptions and which assumptions dominate the result. It does not estimate muscle preservation, discomfort, long-term efficacy, clinical safety, or mission readiness. No human or animal data were collected.

6.2 Model

We sampled an illustrative energy model with triangular distributions for incident irradiance, active area, textile conversion efficiency, optical and packaging losses, MPPT and storage efficiency, electronics overhead, and stimulation demand. For each draw, harvested electrical energy was estimated as

$$E_{\mathrm{harvest}} = G A_{\mathrm{PV}} \eta_{\mathrm{PV}} \tau \eta_{\mathrm{MPPT}} \eta_{\mathrm{store}} t_{\mathrm{charge}}, \tag{2}$$

where G is irradiance, A_{PV} is active area, η_{PV} is conversion efficiency, τ captures optical and packaging transmission, and t_{charge} is usable charging time. The session demand includes a simplified pulsed electrode load and a fixed electronics overhead:

$$E_{\mathrm{demand}} = t_{\mathrm{session}} (P_{\mathrm{electrode}} + P_{\mathrm{overhead}}). \tag{3}$$

The energy adequacy ratio is $\mathrm{EAR} = E_{\mathrm{harvest}} / E_{\mathrm{demand}}$. A draw was counted as a preliminary energy-support case when $\mathrm{EAR} \geq 1.20$ and the estimated output voltage remained below a declared compliance-envelope value. These thresholds are screening rules for the model, not medical or engineering standards.

6.3 Scenarios, protocols, and reproducibility

The probe crossed five scenarios with three candidate protocols and generated 50,000 draws per scenario–protocol pair (750,000 observations total). The scenarios were indoor habitat, window-filtered habitat, lunar deployed sunlight, Mars clear deployed sunlight, and light interruption. The protocols were a 30-minute two-channel bridge-light session, a three-by-30-minute anchored session, and a

45-minute four-channel multichannel screen. These names identify load profiles in the model; they do not prescribe a stimulation regimen.

The run used a fixed seed (20260718), standard-library Python, versioned assumptions, boundary-case tests, and a SHA-256 manifest. An initial post-run audit found a defect in a unit-name checker; the checker was corrected without changing model equations or numeric outputs, and the final verification passed 17 of 17 tests. This audit trail is part of the result because evidence-bound speculation requires visible correction rather than silent cleanup.

6.4 Results

Table 2 reports median EAR and joint screening probability. The full distribution, sensitivity analysis, boundary cases, assumptions, and code are supplied in the accompanying project directory.

Table 2. Selected results from the illustrative computational probe. Values are model outputs, not measured device performance.

Scenario	Protocol	Median EAR	Joint success	Interpretation
Indoor habitat	Bridge-light	1.887	71.9%	Conditional
Indoor habitat	Anchored	0.626	19.1%	PV-assisted only
Window-filtered	Bridge-light	26.639	99.96%	Preliminary support
Lunar deployed	Bridge-light	88.643	99.99%	Preliminary support
Mars clear deployed	Anchored	12.117	99.91%	Preliminary support
Light interruption	Anchored	0.095	0.60%	Autonomy unsupported

Across all 15 scenario–protocol pairs, 9 met the preliminary energy-support rule, 1 was conditional and assumption-sensitive, 4 supported only a PV-assisted framing, and 1 did not support energy autonomy. Deployed or strongly illuminated scenarios were robust under the declared assumptions. Indoor operation was conditional or assisted. Interrupted-light operation was predominantly assisted or unsupported. Irradiance and charge duration were the strongest positive sensitivity drivers; electronics overhead was a recurring negative driver. All zero-light boundary cases failed, and all deliberate voltage-compliance stress cases exceeded the assumed envelope.

The result changes the design argument in a productive way. A photovoltaic garment cannot be specified as simply "solar powered." It must expose a context-dependent state: autonomous support, hybrid support, or unavailable support. The safe response to poor resource conditions is graceful degradation, not a hidden increase in bodily stimulation.

7 Discussion: What the Artifact Makes Debatable

7.1 Resource-to-body legibility

The suit turns an environmental conversion chain into an interaction object. Conventional battery interfaces hide the source of energy, while a photovoltaic

garment makes energy provenance spatial and temporal: one part of the garment may be illuminated while another is shaded; a reserve may have been accumulated hours earlier; and dust or window filtering may introduce uncertainty. Showing provenance can support trust, but it can also burden the wearer with information. A future study should compare simple reserve displays with explanations that expose environmental uncertainty and predicted session duration.

7.2 Bodily sovereignty and automation

Energy-aware automation is attractive because it could schedule charging and recommend a contingency session around crew workload. Bodily actuation is different from ordinary scheduling, however. A system that silently starts or escalates stimulation would convert resource optimization into bodily governance. The proposed boundary is therefore deliberately conservative: automation may predict, summarize, and recommend, but initiation, stop, and envelope changes remain under explicit human or authorized clinical control.

7.3 Mass, maintainability, and infrastructure politics

The garment adds photovoltaic area, storage, electrodes, control electronics, insulation, cleaning, replacement, and verification work. The design is not automatically preferable to a centralized exercise device. Its value depends on whether distributed redundancy justifies these costs under a mission's failure model. This is a mission-architecture question as much as a wearable-design question. The suit should be compared with alternatives such as a spare exercise interface, shared reserve power, passive movement protocols, or a repairable modular stimulator.

7.4 A boundary object for SpaceCHI

The artifact enables productive disagreement across communities. Space-health researchers can ask whether the bounded physiological target is meaningful. Power engineers can challenge the illumination and storage assumptions. Wearable designers can question why harvesting and actuation belong in clothing. Human-factors researchers can test whether provenance and stop states are intelligible. Mission planners can evaluate mass and maintenance. HCI researchers can study how a garment becomes infrastructure. The position paper does not require these communities to agree that the suit is deployable; it asks them to articulate which boundary conditions would make the proposition useful or untenable.

8 Research Agenda and Falsification Conditions

The proposal is intentionally staged. Stage 1 should measure material and electrical feasibility using calibrated plane-of-array irradiance, textile or panel output,

storage losses, full-system Wh per session, thermal behavior, and fault isolation. Stage 2 should characterize human factors in a laboratory with a certified stimulator: fit, contact stability, preview comprehension, discomfort, local stop performance, and user refusal. Stage 3 should use bed-rest or analog studies to test evoked force and short-term countermeasure outcomes under an approved protocol. Stage 4 should evaluate maintainability, dust, thermal cycling, radiation-relevant packaging, and habitat integration. Only after these stages should a flight or analog mission study be considered.

The metabolic-safety-net position would be narrowed or rejected if any of the following hold: (1) the mass and maintenance burden dominate the value of distributed continuity; (2) realistic illumination and storage cannot support even a small bounded session during the target outage; (3) users cannot reliably understand or stop the stimulation; (4) the system's uncertainty makes safe automation impossible; or (5) controlled physiological studies show no meaningful benefit for the selected contingency target. These conditions prevent the artifact from becoming an unfalsifiable product narrative.

9 Limitations

The computational probe uses illustrative distributions rather than calibrated hardware measurements. It simplifies geometry, spectrum, temperature, dust, storage aging, electrode impedance, converter behavior, and protocol scheduling. Its voltage rule is a declared screening boundary, not a certification threshold. The probe does not model human physiology, muscle preservation, discomfort, clinical eligibility, or long-duration mission outcomes. The suit architecture is conceptual; the figures are placeholders until the author supplies an annotated design. Finally, the position is one proposal among many and should not displace established exercise countermeasures or clinical care.

10 Conclusion

The Superman Suit reframes a photovoltaic NMES garment as a question about infrastructure, not enhancement. Its contribution is the metabolic safety net: a bounded, distributed capability that preserves limited physiological care during interruption while keeping energy provenance, scarcity, uncertainty, and bodily agency visible. The design propositions and failure-first probe show why "solar powered" is not a sufficient systems description. Strong illumination may support an autonomous state, indoor operation may be conditional, and interrupted light may make autonomy unavailable. That variability is not a defect to hide; it is the HCI problem. SpaceCHI can help determine when a body-proximate reserve is worth its mass and complexity, how it should be governed, and which evidence would justify moving from a theory-bearing artifact toward a validated countermeasure.

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